

General Comments

The authors depict a Bayesian model and inference algorithm for deconvolutions of potentially noisy and partially observed images. The emphasis is on computation and the authors are exceptionally detailed in the mathematics and practicalities of the algorithm: a reader could quite reasonably implement the method using this manuscript.

However, the innovations over Senn et al. (2025) are not clear. From the Introduction of the manuscript: “*Our contribution is threefold. First, we formulate the model outside its original seismic setting. Second, we derive an explicit eigenvalue-based representation of the posterior computations and Gibbs sampler, so that the main updates can be implemented directly in the spectral domain. Third, we introduce a marginal blur update obtained by analytically integrating out the image from the joint blur-image conditional and sampling the resulting marginal blur conditional with Hamiltonian Monte Carlo (HMC).*”

Substantial movement on this first point is critical, since JCGS is not a geophysical journal. The method needs to be formulated *and applicable* outside of the original seismic application. But the assumptions of 1) a vertical/horizontal separable prior, 2) row-uniform observations, and 3) a 1-dimensional convolution kernel would be prohibitive for any other canonical deconvolution application (Gaussian, defocus, motion blur, such as in the seminal papers cited in the current manuscript, Levin et al., (2009); Babacan et al., (2012); Orieux et al., (2010); Park et al., (2012)). These assumptions are buried within the middle of the manuscript (except for the 1D kernel, which is in the abstract) and are in conflict with the title, “*Bayesian Semi-Blind Deconvolution at Scale*” and the general 2D deconvolution statement in equation 1, which suggest a very general method to a researcher searching the literature. The method is only tested on simulated 1D images (for which all the above restrictions are vacuous) and a seismic dataset with a single well observation. For a statistics journal, the paper would be much improved by relaxing the above restrictions and demonstrating applicability in another domain.

On the second and third contributions, the benefit of Gibbs updates in the spectral domain over Senn et al. (2025) is never explicitly stated, and the collapsed HMC update provides improvements in limited scenarios, by the authors’ admission. But the weight of any improvements on these fronts is entirely conditional of the applicability of the method.

Specific Comments

1. Reading the paper in sequence, I found Section 2.1 confusing and over-engineered, primarily because the reasons behind the formulation (which are good reasons!) are not apparent until much later in the manuscript.

The authors introduce the general linear Gaussian model $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{e}$ and the conditional distribution for $\mathbf{x}|\mathbf{y}$. But later the authors suppose $\mathbf{e} = 0$, $\mathbf{A}$ is a selector matrix (so $\mathbf{A}\mathbf{x} = \mathbf{x}_c$), and $\mathbf{A}\mathbf{x} = \mathbf{b}$ (so $\mathbf{x}_c = \mathbf{b}$). At first read, this raises the question “Why not just sample $\mathbf{x}_u|(\mathbf{x}_c = \mathbf{b})$?”. Quantities $\mathbf{A}$ and $\mathbf{e}$ seem like dead weight, and the dismissed alternative, forming the singular $N(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*)$ and sampling, is a straw man, since the obvious competitor is the non-singular conditional for $\mathbf{x}_u$.

Algorithm 1 is attractive, but not for the reasons given in Section 2.1: the unconstrained covariance is circulant (Sections 3 & 4), so Step 1 of Algorithm 1 is cheap, while subsetting destroys that structure with little reduction in dimension if $k \ll n$. The reason given, that conditioning by Kriging “can be computationally efficient for $k < n$ ”, is not true for a generic matrix, where conditioning by Kriging is actually strictly worse than the partitioned

conditional.

Further, the authors handle two different cases, one where $\mathbf{A}$ is a selector and $\mathbf{e} = \mathbf{0}$ and another where $\mathbf{A}$ is general and $\mathbf{e} \sim N(\mathbf{0}, \mathbf{\Sigma}_e)$. So the general equation is truly warranted, but this is not clear in Section 2.1.

I encourage the authors to clearly delineate the two cases they will deal with and forecast why Algorithm 1 will be preferable to the more obvious $\mathbf{x}_u | (\mathbf{x}_c = \mathbf{b})$ sampler.

2. Page 7, Line 31 : Should be “Hermitian *transpose*”, or conjugate transpose, which is more common.
3. Equations 5 and 9, the conjugate transpose should be on the right $\mathbf{F}_n$. Further, the eigenvalues just below Equation 9 are missing a $\sqrt{mn}$ factor.
4. Page 8, Line 4 and 17 : “eigenvalue-based operations”? I do not think that is a standard term. The listed operations end up only affecting the eigenvalues (because of the shared eigenvectors), but that’s a consequence, not a definition for that set of operations.
5. Section 2.3 : Would the authors please justify their choice of a horizontal-vertical separability? In general, this is a limiting choice for a generic spatial field, and is not required for a circulant embedding (Wood and Chan, 1994, Dietrich and Newsam, 1997). It may be appropriate for the intended geophysical application, however. Also, the order $\mathbf{\Sigma}_h \otimes \mathbf{\Sigma}_v$, assumes the $\mathbf{x}$ lattice is vectorized column-major, which probably should be stated explicitly in the manuscript.
6. Page 11, Line 1: “*We assume that the additional image information consists in m exactly observed pixels at the same rows of a subset of columns in the image*”. It would be best to write this out explicitly and mathematically. But more importantly, this is highly restrictive, and buried in the middle of the paper. Again, perhaps this restriction is fine for the geophysical application (wells within an area tend to be drilled to a standardized depth?), but it would be prohibitive elsewhere.
7. Page 11, Paragraph 2: Again, a 1D stationary blur kernel is very restrictive...